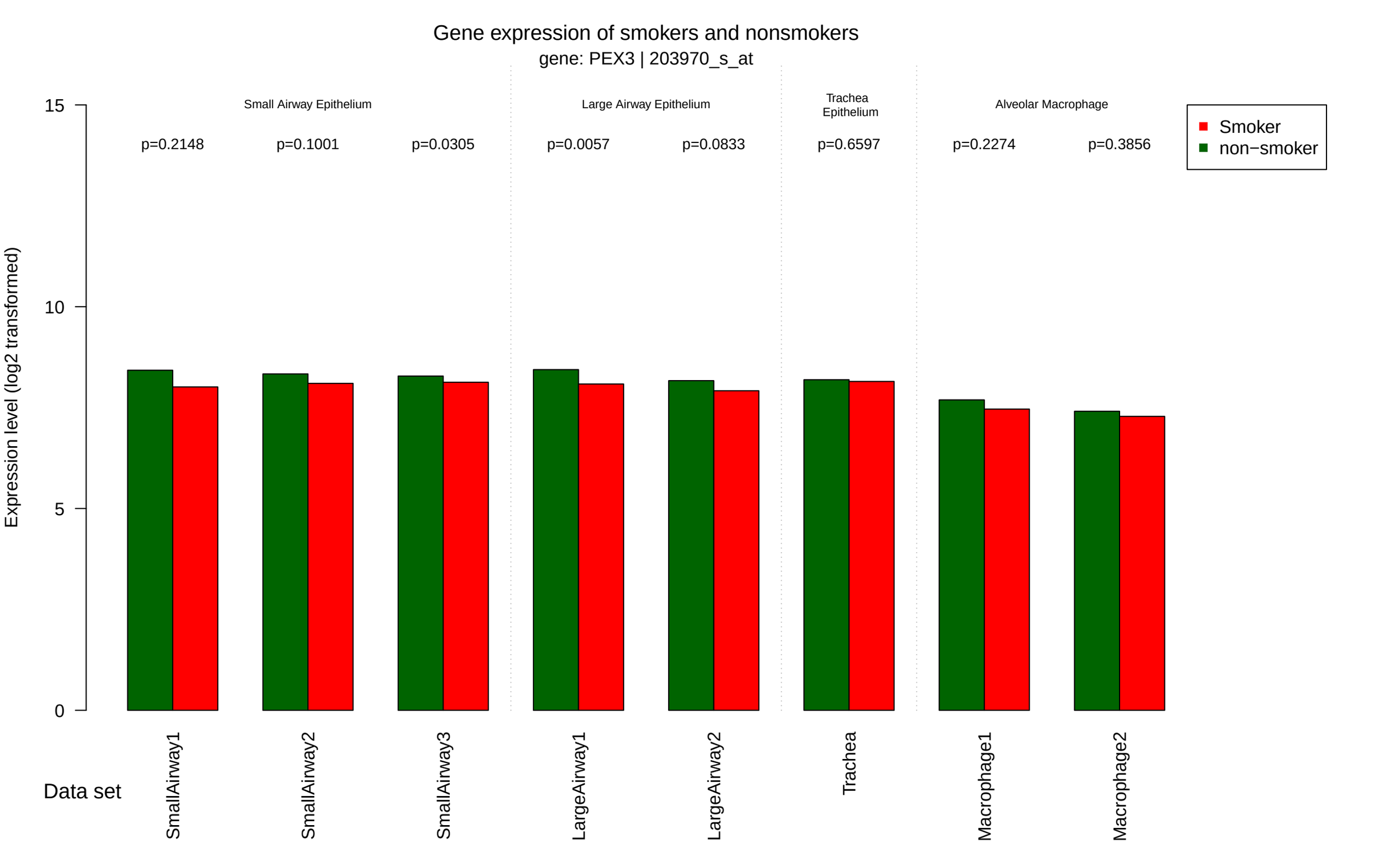
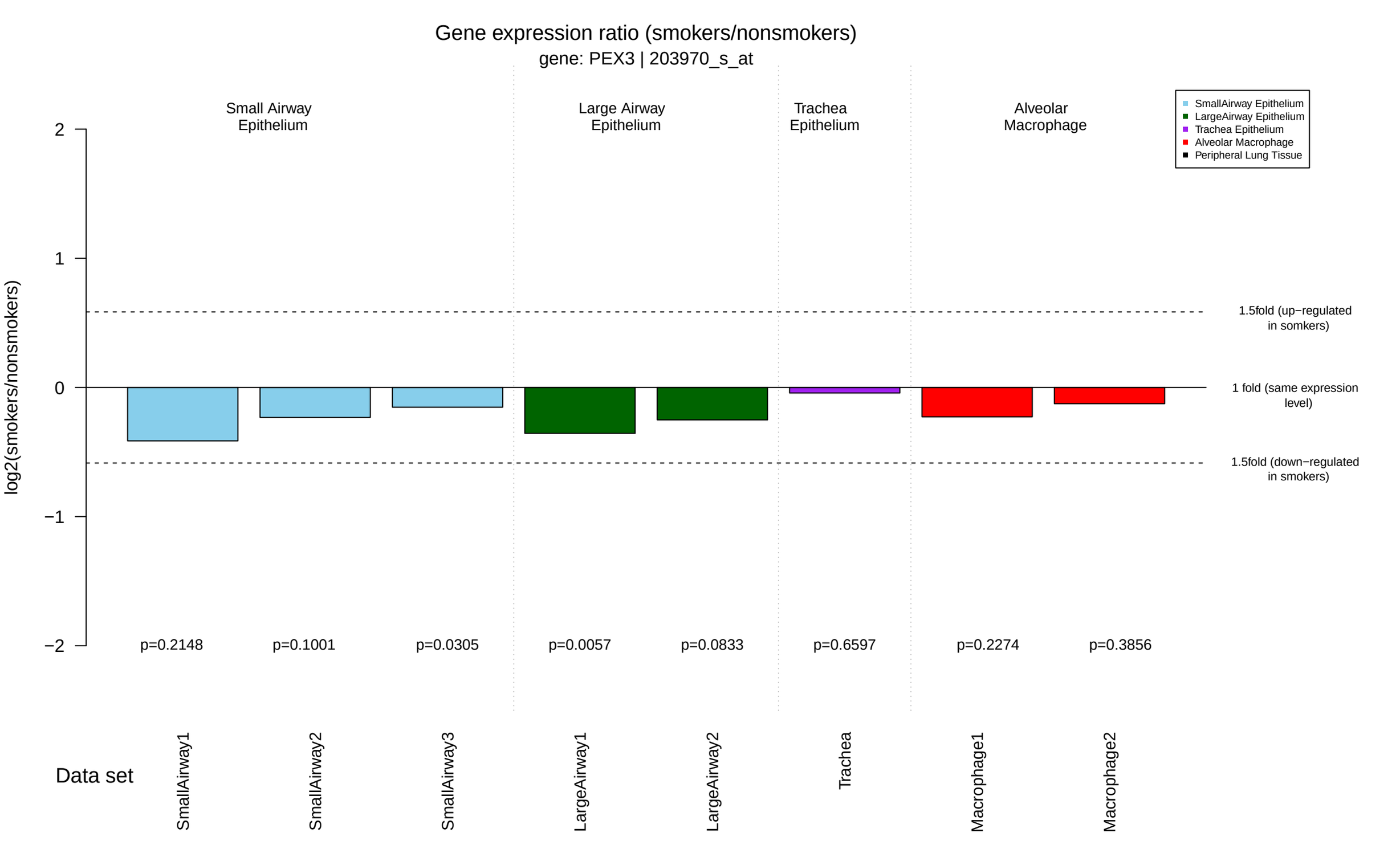
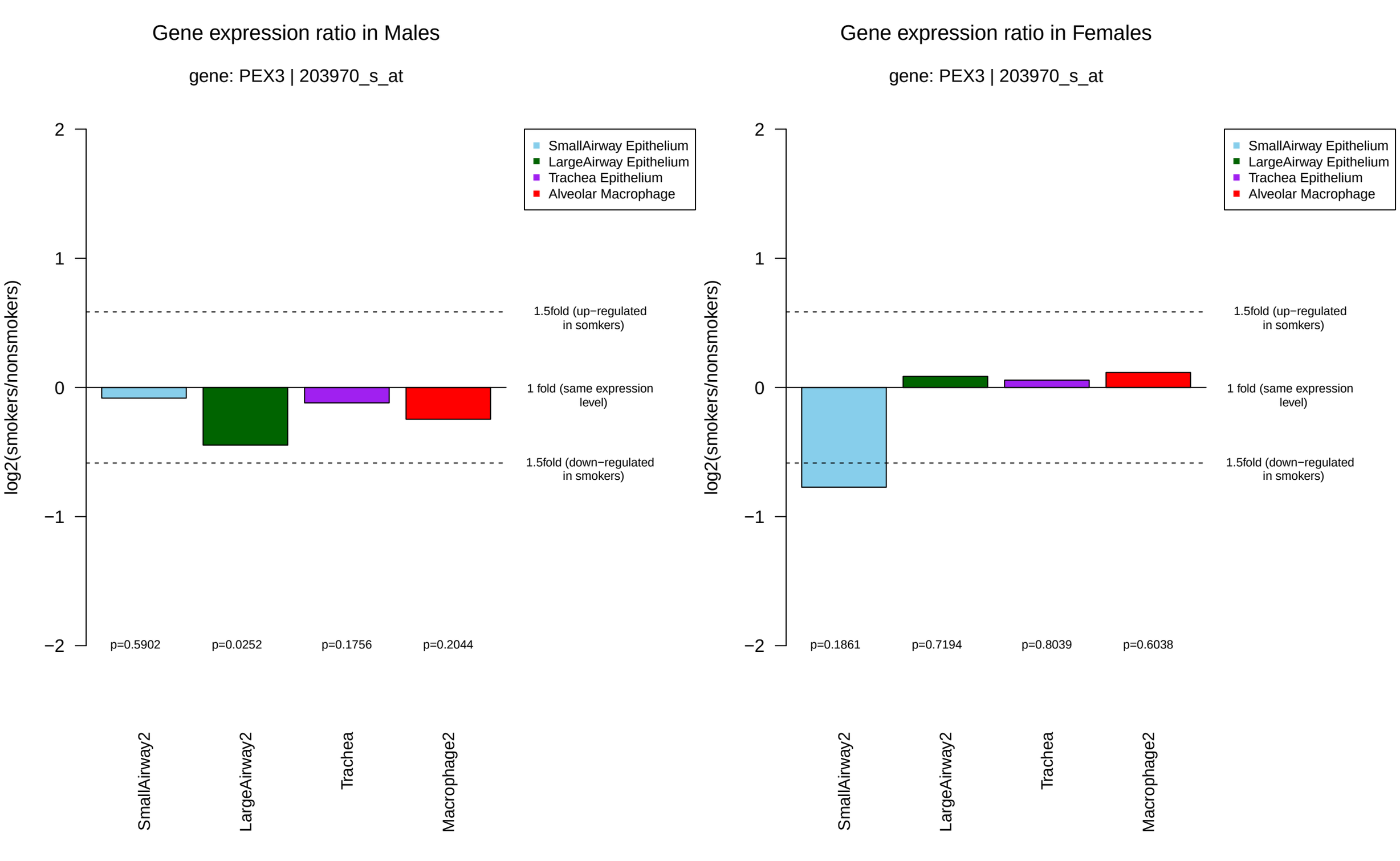
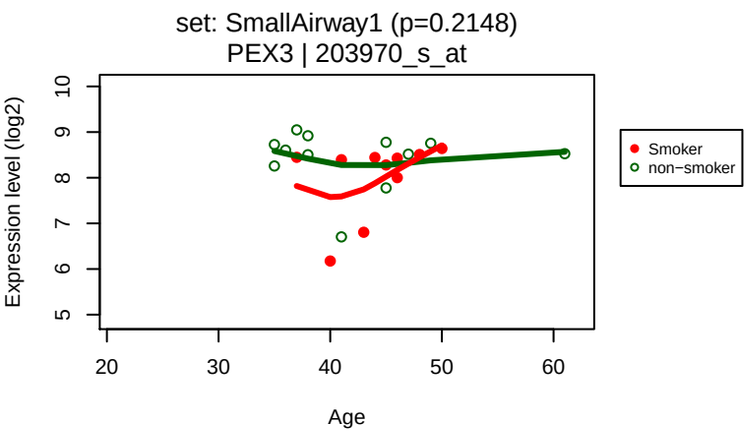
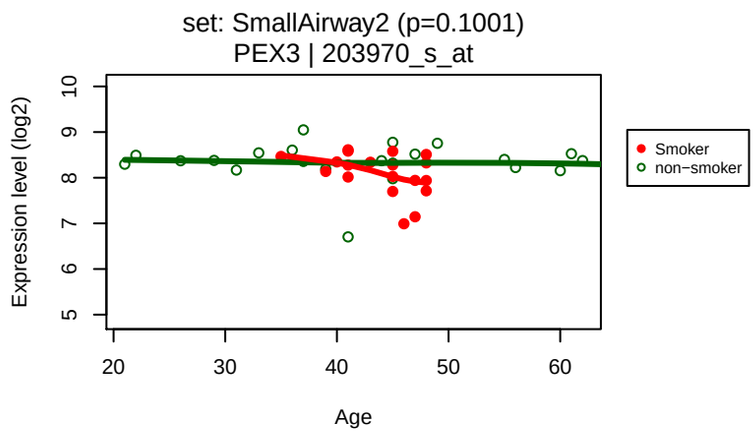
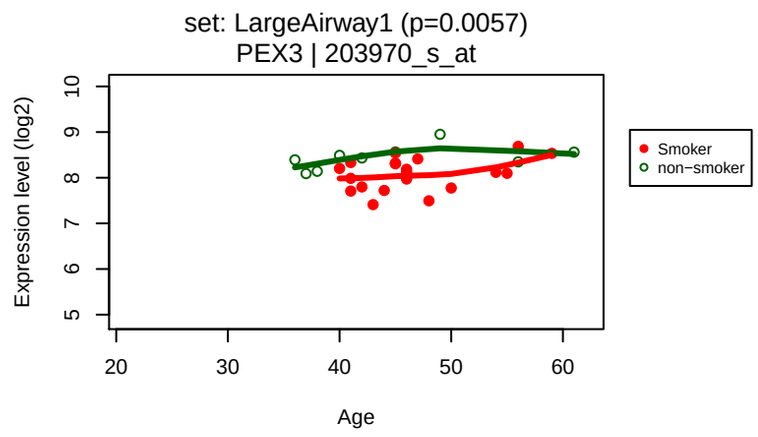

Gene expression of smokers and nonsmokers
gene: PEX3 | 203970_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2148
p=0.1001
p=0.0305
p=0.0057
p=0.0833
p=0.6597
p=0.2274
p=0.3856
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

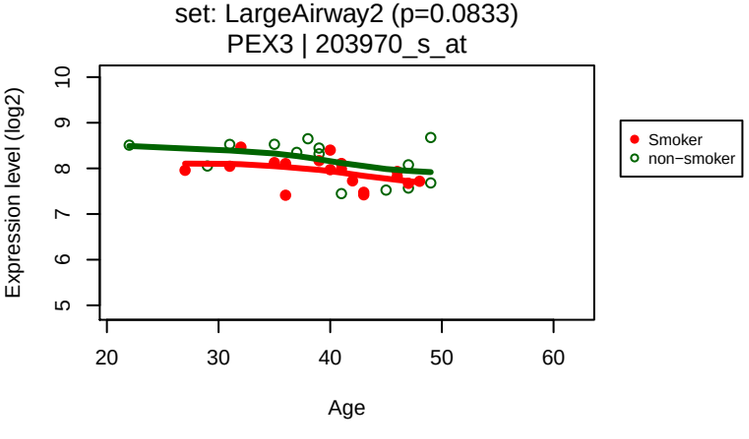

Gene expression ratio (smokers/nonsmokers)
gene: PEX3 | 203970_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2148
p=0.1001
p=0.0305
p=0.0057
p=0.0833
p=0.6597
p=0.2274
p=0.3856
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

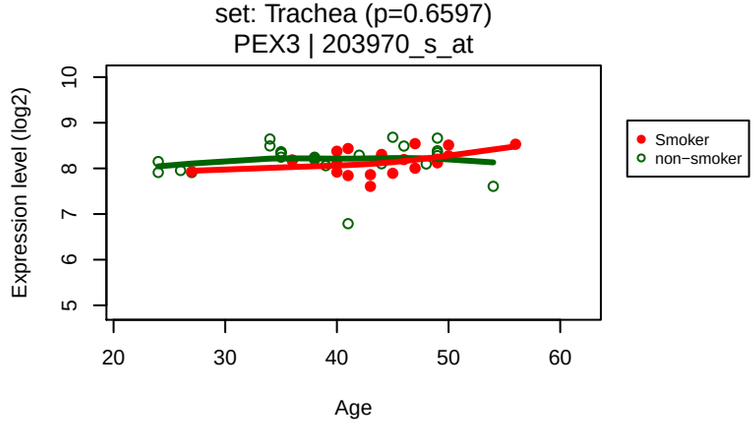

Gene expression ratio in Males
gene: PEX3 | 203970_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5902
p=0.0252
p=0.1756
p=0.2044
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PEX3 | 203970_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1861
p=0.7194
p=0.8039
p=0.6038
SmallAirway2
LargeAirway2
Trachea
Macrophage2

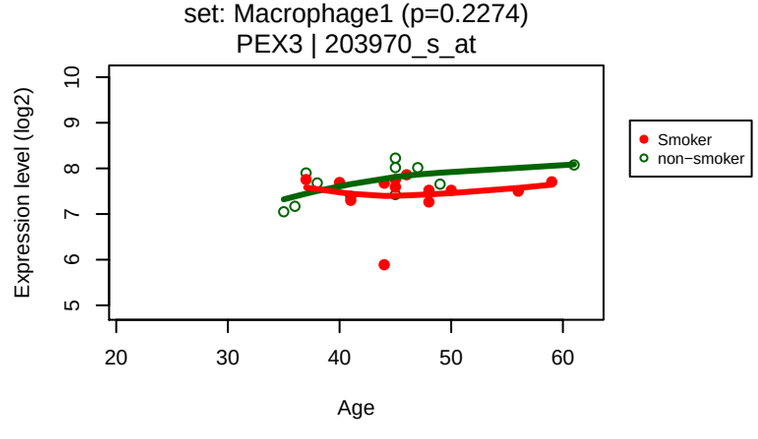

set: SmallAirway1 (p=0.2148)
PEX3 | 203970_s_at
Expression level (log2)
Age
Smoker
non−smoker

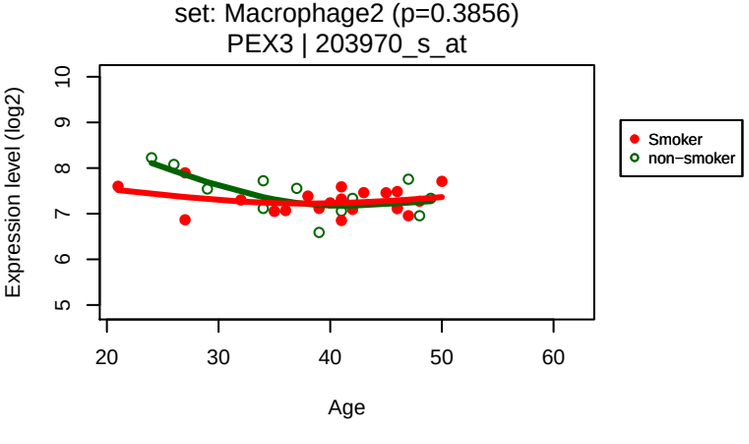

set: SmallAirway2 (p=0.1001)
PEX3 | 203970_s_at
Expression level (log2)
Age
Smoker
non−smoker

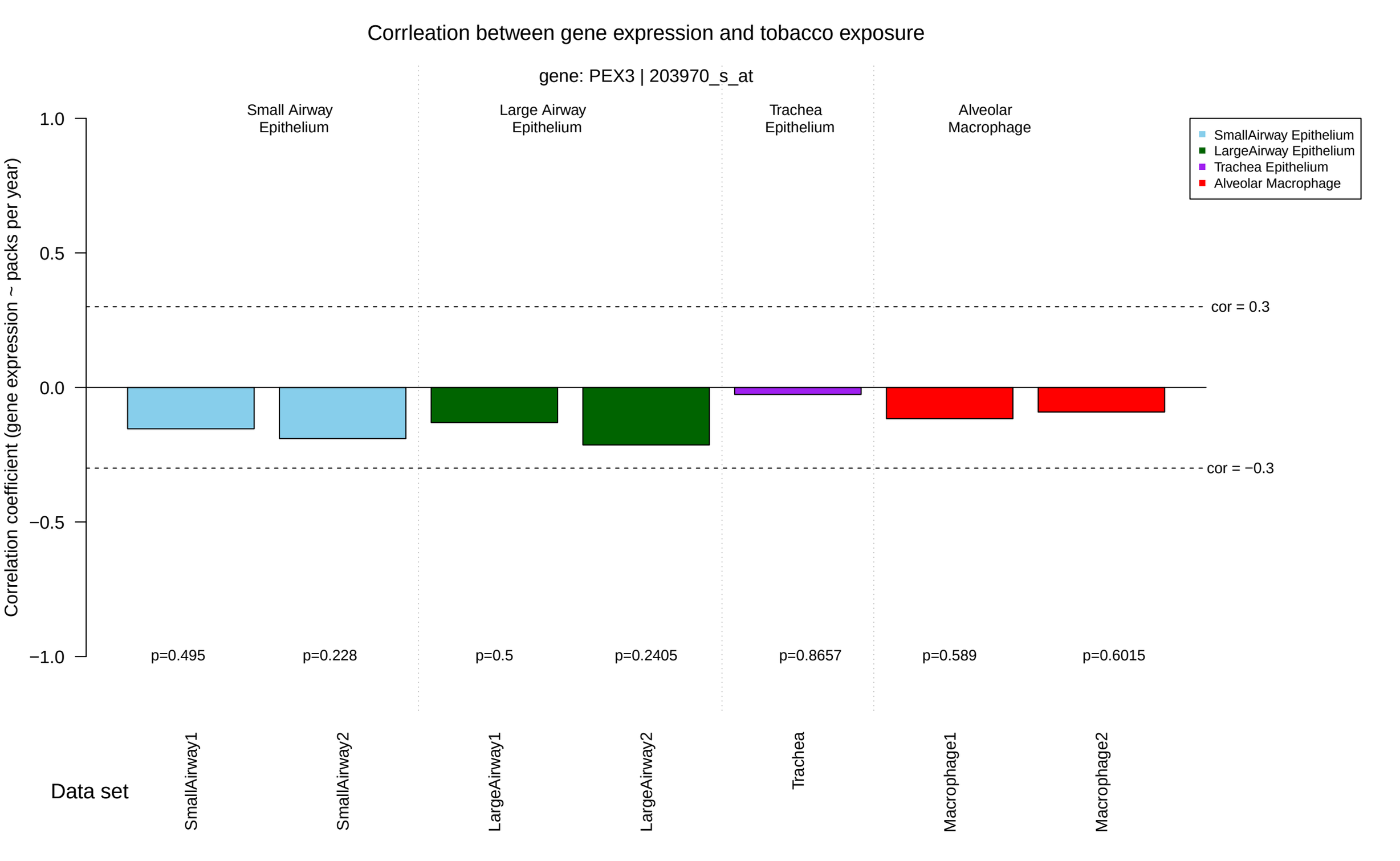

set: LargeAirway1 (p=0.0057)
PEX3 | 203970_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0833)
PEX3 | 203970_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6597)
PEX3 | 203970_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2274)
PEX3 | 203970_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3856)
PEX3 | 203970_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PEX3 | 203970_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.495
p=0.228
p=0.5
p=0.2405
p=0.8657
p=0.589
p=0.6015
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2